

Name _____ Period _____

Skill 31.1 Exercise 1	
label each pair of events as independent or dependent and give a one-sentence reason	
A coin is flipped twice. Event A: first flip is heads. Event B: second flip is heads.	
A card is drawn from a deck, <u>not</u> replaced, then a second card is drawn. Event A: first card is a heart. Event B: second card is a heart	
A card is drawn, replaced, shuffled, then a second card is drawn. Event A: first card is a heart. Event B: second card is a heart.	
A spinner with equal sections is spun twice. Event A: first spin lands on blue. Event B: second spin lands on blue.	
A bag has 4 green and 6 yellow marbles. Two marbles are drawn without replacement. Event A: first marble is green. Event B: second marble is green.	

Skill 31.2 Exercise 1	
label each pair of events as mutually exclusive or not mutually exclusive and give a short reason.	
Event A: roll an even number or Event B: roll an odd number	
Event A: roll a number greater than 4 or Event B: roll an even number	
Event A: roll a 2 or Event B: roll a number less than 4	
Event A: roll a prime number or Event B: roll an odd number	

Name _____ Period _____

Skill 31.3 Exercise 1	
Calculate the probability of rolling a prime number or rolling an odd number by completing the steps.	
Step 1: Is this a mutually exclusive or not mutually exclusive?	
Step 2: What is $P(A)$?	
Step 3: What is $P(B)$?	
Step 4: What is $P(A \text{ and } B)$?	
Step 5: What is $P(A \text{ or } B)$?	

Skill 31.4 Exercise 1
A bag contains 3 red marbles and 2 blue marbles . Two marbles are drawn without replacement .
Write a conditional probability expression (using notation) for each statement:
a. The probability that the second marble is red , given that the first marble is blue
b. The probability that the second marble is blue , given that the first marble is red
For each statement above, calculate the probability.

Skill 31.4 Exercise 2
A bag contains 3 red marbles and 2 blue marbles . Two marbles are drawn with replacement .
Write a conditional probability expression (using notation) for each statement:
a. The probability that the second marble is red , given that the first marble is blue
b. The probability that the second marble is blue , given that the first marble is red
For each statement above, calculate the probability.

Name _____ Period _____

Skill 31.5 Exercise 1

A bag has 3 red and 2 blue marbles. Two marbles are drawn **without replacement**. Write an expression for: *The probability the first is red **and** the second is blue*

Compute marble probabilities (without replacement). Using the same bag (3 red, 2 blue), two draws without replacement.

Calculate:

- $P(\text{Blue 1st and Blue 2nd})$
- $P(\text{Blue 1st and Red 2nd})$

Compute the probabilities with replacement using the same bag.

Name _____ Period _____

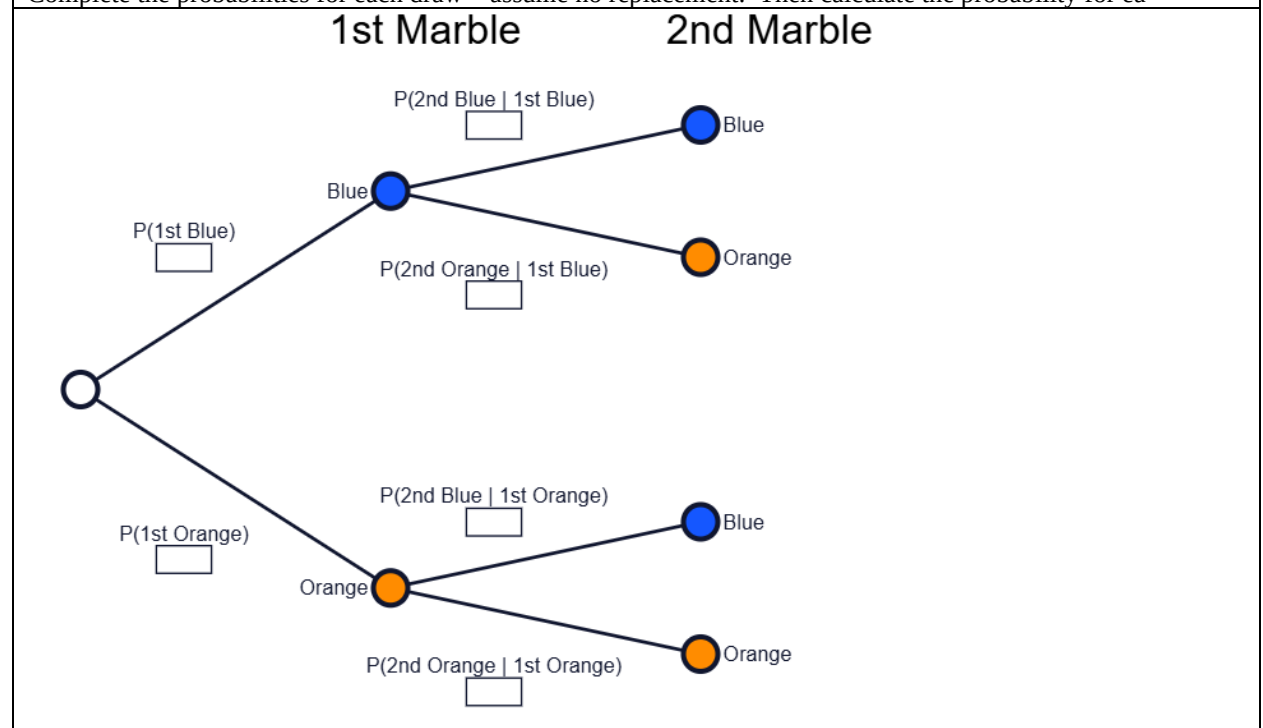
Skill 31.6 Exercise 1

Below is a two-draw probability tree for drawing marbles from a bag.

Given the distribution of blue and orange marbles below,



Complete the probabilities for each draw – assume no replacement. Then calculate the probability for ea



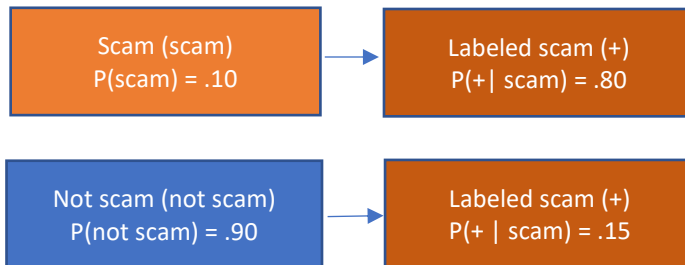
Name _____ Period _____

Skill 31.7 Exercise 1

Your phone uses a filter to label text messages as “Scam” or “Not Scam.” The filter is helpful, but it is not perfect. Suppose the following are true,

- 10% of all messages are actually scams ($P(\text{scam}) = .10$)
- 90% of all messages are not scams ($P(\text{not scam}) = .90$)
- If a message is a scam, the filter labels it a scam 80% of the time ($P(\text{label} | \text{scam}) = .80$)
- If a message is not a scam, the filter still labels it as a scam 15% of the time ($P(\text{label} | \text{not scam}) = .15$)

These outcomes are shown visually below.



If your phone labels a message as “Scam,” what is the probability that the message is actually a scam? This is, $P(\text{scam} | +)$

Consider 100 messages. Based on the probabilities above, how many are scams? How many are not scams?

Based on the probabilities above, how many of the *scams* are correctly labeled *scam*?

Based on the probabilities above, how many of the *not scams* are incorrectly labeled as *scam*?

What is the total messages that are labeled as *scam*?

What is $P(\text{scam} | +)$. That is, out of the messages labeled as scam (+), how many were actually scams?